

Public Economics

Lecture 6 · Inequality

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Today's Questions

- What exactly do we mean by “inequality,” and why do economists — and everyone else — care about it?
- How do we measure inequality in income and in wealth, and what are the pitfalls of each approach?
- How much of a person's economic position is inherited from their parents, rather than earned?

Based on

Leenders, W., *Microeconomics of Public Policy, Part III: Inequality* (Sciences Po, 2023–2024).

6.1 What Is Inequality, and Why Do We Care?

6.2 Measuring Inequality

6.3 Income Inequality

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6.1 What Is Inequality, and Why Do We Care?

What Is Inequality?

Inequality refers to differences in resources between individuals or groups.

- **Type of resource:** income, wealth, consumption, “opportunity.”
- **Type of group:** class, gender, region, generation, ...
- **Scope:** global, national, regional.

The World Inequality Lab tracks the most up-to-date research in the *World Inequality Report* and at wid.world.

Income and Wealth Shares Around the World

In every world region, the top 10% earns a far larger share of income than the bottom 50% — and the gap is wider still for wealth, where the top 10% typically owns 60–80% and the bottom half less than 5%.

Region (2021)	Bottom 50%	Middle 40%	Top 10%
Europe	19%	45%	36%
East Asia	14%	42%	43%
North America	13%	41%	46%
MENA	8%	32%	58%

Europe is the world's most equal region by this measure — but still far from equal.

Source: World Inequality Report 2022.

Why Do We Care?

- **Eurobarometer:** 81% of EU citizens agree that differences in income are too great.
- Biologists have documented an aversion to inequality among capuchin monkeys, apes, dogs, and birds (Brosnan & de Waal, 2014) — a sense of fairness may be evolutionarily old.
- Humans engage in joint production with others through families, firms, and the state. The fruits of that production need to be divided somehow — which may explain why fairness concerns run so deep.

⇒ Emmanuel Saez's 2021 AEA Distinguished Lecture, *"Public Economics and Inequality: Uncovering Our Social Nature."*

6.2 Measuring Inequality

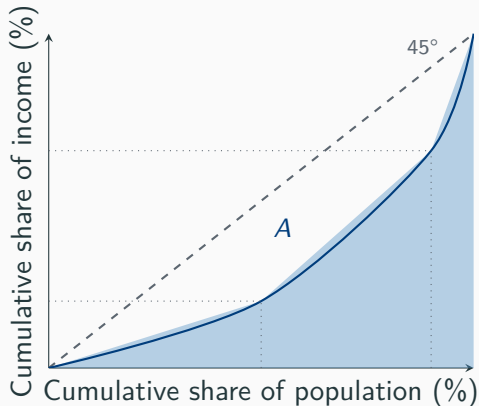
Income Shares and Summary Indices

- Inequality is distinct from **poverty**: it concerns the *relative* distribution of resources, not an absolute threshold.
- **Income shares** measure how much of total income accrues to a given group — the bottom 50%, the top 10%, the top 0.01%.
- **Indices** compress the whole distribution into a single number: the Gini coefficient, the Atkinson index, the Theil index.

The cost of simplicity

A single number is easy to compare across countries and years — but it collapses a complex, multidimensional concept, and (changes in) it are often hard for non-experts to interpret (Piketty, *Capital in the 21st Century*, pp. 266–267).

The Gini Coefficient



- Plot the **Lorenz curve**: the cumulative share of income earned by the bottom $x\%$ of the population.
- Perfect equality is the 45° line; the further the Lorenz curve sags below it, the more unequal the distribution.
- **Gini** = $2A$, where A is the area between the two curves — ranges from 0 (perfect equality) to 1 (perfect inequality).

Data on Inequality: Surveys vs. Tax Records

- **Household surveys** are the traditional source — but the sample is often too small to say much about the top, incomes/wealth above a threshold are often top-coded, and very few surveys existed before the 1950s.
- **Tax data** (Pareto, Kuznets, Piketty–Saez tradition) reach further into the top of the distribution, but are distorted by tax evasion, differ across countries' tax laws, and don't map cleanly onto macroeconomic income concepts (GDP, national income).

Distributional national accounts (DINA)

A newer approach reconciles survey and tax data with macroeconomic aggregates, using a consistent, comprehensive income definition — measured both before and after taxes and transfers. Austria has its own DINA series (Jestl & List, 2020, 2004–2016) — recall Lecture 1.

6.3 Income Inequality

What Is Income?

National income = domestic output, net of capital depreciation, plus net foreign income — made up of **labor income** ($\approx 70\text{--}75\%$) and **capital income** ($\approx 25\text{--}30\%$).

Labor income (wages, pensions, fringe benefits) inequality comes from:

- Education and skills
- Labor-market institutions (unions, minimum wages)
- Work effort
- Social norms and discrimination

Capital income (profits, interest, rent) inequality comes from:

- Wealth inequality itself
- Inequality in *rates of return* on that wealth

Pre-Tax and Post-Tax Inequality

Governments redistribute income through taxation (Lecture 4) and government spending:

- **Cash transfers:** pensions, unemployment benefits, social assistance.
- **In-kind transfers:** goods/services provided free or below cost (healthcare, education).
- **Collective expenditure:** non-rival spending consumed passively by everyone (defense, infrastructure).

Government spending runs 30–50% of GDP across rich democracies — so the gap between *pre-tax* and *post-tax* inequality can be large.

Predistribution, Not Just Redistribution

Top 1% income share	Pre-tax		Post-tax	
	1980	2017	1980	2017
United States	11%	21%	9%	16%
Europe	8%	11%	7%	9%

- Blanchet, Chancel & Gethin (2022) construct distributional national accounts for 26 European countries, 1980–2017.
- Europe redistributes a *larger* fraction of output overall — but US redistribution is more targeted at the bottom 50%.
- **Predistribution** (a more equal distribution *before* taxes and transfers) — not redistribution — is the main reason Europe ends up more equal than the US.

Source: Blanchet, Chancel & Gethin, “Why Is Europe More Equal than the United States?” *AEJ: Applied Economics* 14(4), 2022.

Austria in the OECD Income Distribution

	Bottom	2nd	3rd	4th	Highest	Top 10%
Sweden	10.7%	14.4%	17.6%	21.5%	35.7%	10.9%
Austria	8.4%	12.4%	16.8%	22.3%	40.1%	13.6%
France	9.4%	12.9%	16.3%	21.0%	40.4%	15.2%
United States	3.3%	8.5%	14.6%	23.4%	50.2%	21.3%
OECD average	8.5%	12.2%	16.0%	21.1%	42.2%	16.7%

Austria sits close to the Nordic end of the distribution — more equal than the OECD average, and far more equal than the US.

Source: Gruber, *Public Finance and Public Policy*, §17.1, income-quintile shares by country.

Austria's Gini Coefficient: Stable, Then Rising

- Austria's most recent income Gini coefficient (equivalised disposable household income) is **28.4** (EU-SILC 2024 release, income year 2023) — just below the EU-27 average of **29.4**.
- Halla & Weber (2024) find Austria's income inequality was **broadly stable from 2004–2019**, despite major labor-market shifts over that period: rising female labor-force participation, educational upgrading, and growth in part-time and cross-border work.
- More recently, Statistik Austria reports the Gini **rose for three consecutive years** through 2024 — the highest level in the EU-SILC series to date.

A low, stable headline number can still mask a recent turning point — always check whether “stable” inequality is still stable in the latest data.

Source: Eurostat, EU-SILC (*ilc_di12*); Halla & Weber, “Persistent Low Inequality Despite Compositional Shifts in Austria,” *Fiscal Studies* 45(3), 2024; Statistik Austria, Einkommen und Armut 2024.

Real-Time Inequality

- The World Inequality Lab's realetimeinequality.org combines national accounts, tax, and survey *microdata* to estimate income growth by group, updated quarterly — rather than waiting years for the next official release.
- Recent readings consistently show income growth **rising with income rank**: the top 0.01% grows fastest, the bottom 50% slowest.

Real-time measurement is itself a research frontier — most official inequality statistics still arrive with a multi-year lag.

Source: Blanchet, Saez & Zucman, “Real-Time Inequality,” working paper, 2022.

6.4 Wealth Inequality

What Is Wealth?

Wealth = the sum of all assets minus the sum of all liabilities.

- An **asset** is a store of value representing expected future economic benefits: real estate, equity, bonds, deposits.
- A **liability** is the converse: mortgages, student loans, consumer debt.

Wealth can be owned by private individuals *or* by the government — which is why **public** wealth matters for inequality too, not just private wealth.

The Rise of Private Wealth, the Fall of Public Wealth

- Across rich countries since 1970, **private** wealth has risen sharply relative to national income — while **public** wealth has stagnated or turned negative.
- Austria's own version of the public-wealth story: the abolition of the general **wealth tax** (*Vermögensteuer*) in 1994, and the lapse of the **inheritance/gift tax** in 2008 (Lecture 4), removed two of the main tools governments elsewhere still use to tax accumulated private wealth.

A government that owns less (or owes more) has less room to cushion shocks, or to redistribute, without new borrowing or taxation.

Source: Blanchet & Martínez-Toledano, "Wealth Inequality Dynamics in Europe and the United States," *Journal of Monetary Economics* 133, 2023.

How Do We Measure Wealth?

- The ideal source would be a well-enforced, comprehensive **wealth tax** — but few countries levy one (Austria included, since 1994).
- In its absence, **surveys** like the ECB's *Household Finance and Consumption Survey* (HFCS) — which includes Austria, run by the OeNB — collect wealth data directly.
- **Inheritance tax records** offer an old alternative: treat estates as a (non-random) sample of wealth and correct for the selection.
- Capital income can be “**capitalised**”: $Y_k = rK \Rightarrow K = Y_k/r$, inferring wealth from the capital income it generates.
- **Rich lists** (Forbes 400, Bloomberg Billionaires Index) compile journalist estimates of the very top.

Austria's Wealth Concentration: A Measurement Case Study

- Austria's HFCS 2021 wave (OeNB) puts the net-wealth Gini coefficient at ≈ 0.70 — and the top 10% of households hold $\approx 51\%$ of net wealth, among the highest concentrations in the euro area.
- The raw survey data put the top 1% share at $\approx 25\%$ — but HFCS-style surveys systematically **undersample the ultra-wealthy** (too few billionaires answer a household survey).
- Correcting for this with rich-list-calibrated methods, OeNB researchers Fessler & Schürz estimate Austria's true top 1% share is closer to $\approx 43\%$.

This is the measurement problem from two slides ago, playing out in real data: the “right” answer for Austria's wealth concentration nearly doubles once you correct for who a household survey misses.

Source: OeNB, Eurosystem HFCS 2021 – First Results for Austria (2023); Fessler & Schürz, “The Size and Distribution of Augmented Wealth,” *Journal of Pension Economics & Finance*, 2026.

What Drives Wealth Inequality?

Wealth owned by individual i in period $t + 1$ depends on wealth in period t , the saving rate, and the rate of return:

$$W_{t+1}^i = (1 + r_t^i) (W_t^i + s_t^i Y_t^i)$$

Differences in wealth can therefore come from:

1. Pre-existing differences in wealth
2. Labor income inequality
3. Differential saving rates
4. Differential rates of return r_t^i

Wealth Concentration: Europe vs. the US

Top 1% wealth share	1970	2020
United States	27%	35%
Europe	24%	24%

- Blanchet & Martínez-Toledano (2023) construct distributional wealth accounts for 31 European countries, 1970–2020.
- Wealth concentration has risen in both Europe and the US recently — but far more in the US.
- Two explanations: European house prices rose by *more* (which spreads gains toward middle-wealth homeowners), and European labor-income inequality rose by *less*.

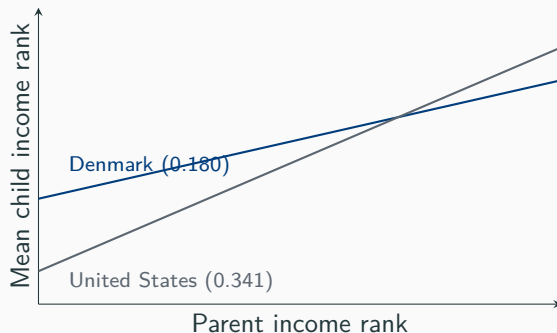
6.5 Intergenerational Mobility

How Do We Measure Mobility?

How much does parental income shape a child's own economic success — a common (if imperfect) proxy for “inequality of opportunity”?

- **Intergenerational elasticity of income:** by how much does a child's income rise if their parents' income rises by 1%?
- **Rank-rank correlation:** the correlation between a parent's income rank and their child's.
- **Absolute mobility:** the expected income rank of a child born to parents at percentile p .

The US Is Less Mobile than Denmark



A flatter slope means a child's income is less tied to their parents' — **more** mobility. The US rank-rank slope (0.341) is nearly double Denmark's (0.180).

Source: Chetty, Hendren, Kline & Saez, "Where Is the Land of Opportunity?" *QJE* 129(4), 2014.

Mobility Varies Enormously *Within* Countries Too

- Chetty et al. (2014) map absolute upward mobility across every US commuting zone using anonymized tax records — the interactive **Opportunity Atlas** lets users explore outcomes down to the neighborhood level.
- Mobility is systematically **lower** in the US South, and higher in the Midwest, Great Plains, and much of the Northeast — differences as large as the US-vs-Denmark gap on the previous slide, found *within a single country*.

The lesson generalizes: national averages can hide as much as they reveal — the same is likely true of mobility across Austria's own regions, even without an Opportunity-Atlas-style tool to show it directly.

Source: Chetty, Hendren, Kline & Saez (2014); opportunityinsights.org.

Austria's Own Mobility Paradox

- Despite its comparatively low income inequality, the OECD finds Austria's **intergenerational mobility lags behind many other OECD countries** — a father's earnings are an unusually strong predictor of his children's adult earnings.
- Schnetzer & Altzinger (2013) estimate an intergenerational income *correlation* of ≈ 0.13 for Austria (EU-SILC data) — European mid-range: more mobile than Southern Europe, less than the Nordics. (A correlation, not the elasticity used in the US/Denmark slopes above — not directly comparable.)

A gap worth naming

The OECD's cross-country "generations to reach the average income" measure (previous slides) has no published Austria-specific figure in the underlying table — even rich-country statistical infrastructure has real gaps.

Source: OECD, *Promoting Social Mobility in Austria* (2020); Schnetzer & Altzinger, *Momentum Quarterly* 2(3), 2013.

Conclusion

Highlights

- Inequality is multidimensional — income vs. wealth, pre-tax vs. post-tax, national vs. within-country — and no single index captures all of it.
- Europe is more equal than the US mainly through **predistribution** (a more equal starting distribution), not just heavier redistribution.
- Wealth is far more concentrated than income everywhere, and driven by a compounding process: existing wealth, saving rates, and rates of return all interact.
- Austria has neither a wealth tax nor an inheritance tax — an unusually light institutional touch on accumulated wealth, even by European standards.
- Intergenerational mobility differs sharply across countries (US vs. Denmark) — and just as sharply *within* them.

Discussion Questions for Next Time

1. If predistribution matters more than redistribution for Europe's relative equality, what pre-tax institutions (labor-market, educational) should get more attention than they currently do?
2. Austria abolished its wealth and inheritance taxes decades ago. Given what drives wealth inequality (the formula on the wealth-inequality slide), what would reintroducing one actually change?
3. Rank-rank mobility in the US is roughly double that of Denmark. What Danish institutions — education, labor market, welfare state — are the most plausible explanations?
4. Should Austria build its own “Opportunity Atlas”? What would it take, data-wise, and what might it reveal about mobility across Austria's own Bundesländer?

- Leenders, W., *Microeconomics of Public Policy, Part III: Inequality*, lecture slides, Sciences Po (MPA 2023–2024).
- Chancel, Piketty, Saez & Zucman, *World Inequality Report 2022*, Harvard University Press.
- Blanchet, Chancel & Gethin, “Why Is Europe More Equal than the United States?” *AEJ: Applied Economics* 14(4), 2022.
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- Chetty, Hendren, Kline & Saez, “Where Is the Land of Opportunity? The Geography of Intergenerational Mobility in the United States,” *QJE* 129(4), 2014.
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- Brosnan & de Waal, “Evolution of Responses to (Un)fairness,” *Science* 346, 2014.
- Piketty, T., *Capital in the 21st Century*, Harvard University Press, 2014.
- Gruber, J. *Public Finance and Public Policy*, §17.1.

Thank you!

Questions and discussion